1- Data Formatting & Handling Missing Values

Removing incomplete records and converting percentage strings into workable decimals.

	نسبة تقدم القدرات (%)	نسبة تقدم التحصيلي (%)
0	-2.63	6.59
1	1.52	3.47
2	-2.80	2.08
3	14.71	-8.04
4	-4.23	7.69

2- Feature Scaling (Normalization)

Compressing numerical features to a 0-1 scale using MinMaxScaler

	قدرات_1447	تحصيلي_1447	التقويم_الخارجي_1447	التقويم_الذاتي_1447
0	0.542109	0.572114	0.521900	0.657232
1	0.452769	0.473143	0.382069	0.470818
2	0.573407	0.656457	0.605593	0.686707
3	0.691388	0.449600	0.520461	0.490142
4	0.498483	0.692800	0.569402	0.550264

3- Linear Regression

Splitting data and training a linear model to extract feature weights

```
--- نتائج نموذج (Multiple Linear Regression) ---
نقطة التقاطع (w0): 0.0505
وزن القدرات (w1): -0.4194
وزن التقويم الخارجي (w2): 0.7999
وزن التقويم الذاتي (w3): 0.4827
```

4- Random Forest

Training a Random Forest model to calculate feature importance scores

```
--- أهمية العوامل في (Random Forest) ---
0.1405 :قدرات_1447
0.6787 :التقويم_الخارجي_1447
0.1807 :التقويم_الذاتي_1447
```

5- XGBoost

Training an XGBoost model: the trees are built one after another, and each new tree learns to correct the errors left by the trees before it.

```
--- أهمية العوامل في (XGBoost) ---
0.0719 :1447_قدرات
0.7516 :1447_التقويم_الخارجي
0.1765 :1447_التقويم_الذاتي
```

6- Model Comparison

Evaluating the three models on the same test set (the 20% of schools never seen in training), then with 5-fold cross-validation for a fairer average. MAE and RMSE are converted back to Tahsili score points.

	Model	MAE (points)	RMSE (points)	R2 (test set)	R2 (5-fold CV)
0	Linear Regression	2.913	3.633	0.666	0.677
1	Random Forest	3.069	3.832	0.628	0.633
2	XGBoost	2.960	3.694	0.654	0.669